Dynamical Generation of the Relaxation Parameter in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) is a one-parameter, CPT-symmetric framework in which all observable deviations from ΛCDM arise from the infrared value of the relaxation parameter α_{IR}. While Companions I–V established the gravitational foundations, tensor phenomenology, relaxation-induced scalar sector, global CPT state, and the first prototype global fit, the microscopic time evolution of the relaxation parameter has not yet been derived on a realistic cosmological background.

This paper, Companion VI, closes this gap. We derive the relaxation equation $\dot{\alpha} = \Gamma\alpha$ from the dimension-six inter-sector operator, show that the naive EFT estimate $\Gamma \sim H^5/M^4$ fails to generate any observable relaxation, and introduce the minimal generalized rate $\Gamma = \beta H$ consistent with the CPT-symmetric structure. Matching to the observed value $\alpha_{\mathrm{IR}} \simeq 0.0175$ fixes $\beta \simeq 1.25 \times 10^{-4}$.

We solve the full evolution equation $\mathrm{d}\alpha/\mathrm{d}a = (\Gamma/H)\alpha$ on a ΛCDM background and find that $\alpha(a)$ grows sharply near the bounce and freezes to its infrared value long before matter–radiation equality. This dynamical behavior explains why the RDCC prediction vector remains stable across more than ten orders of magnitude in scale and provides a natural microscopic origin for the observed value of α_{IR}.

Companion VI thereby completes the dynamical foundation of RDCC and prepares the ground for a full Boltzmann implementation in Companion VII.

Contents

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) describes the Universe as a globally CPT-symmetric bipartite quantum system whose observable deviations from ΛCDM are governed by a single infrared parameter, the relaxation strength $\alpha_{\rm IR} \sim 10^{-2}$. This parameter originates from the unique dimension-six operator

$$\mathcal{L}_{\rm int} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu},$$

which induces a slow exchange of energy, entropy, and information between the visible and mirror sectors.

In Companions I–V, α was treated as an effectively constant infrared quantity that approaches zero at the bounce and flows to its fixed-point value at late times. However, the microscopic time evolution of $\alpha(a)$ has not yet been solved on a realistic cosmological background. This missing step is essential for completing the internal consistency of RDCC: the one-parameter prediction vector

$$\mathcal{O}_{\rm RDCC} = (n_s, \Delta N_{\rm eff}, f\sigma_8, \Omega_{\rm GW})$$

assumes that α is effectively frozen throughout the observable Universe.

The purpose of this paper, Companion VI, is to derive and solve the relaxation equation, quantify the failure of the naive EFT estimate, introduce the minimal generalized rate consistent with the CPT-symmetric structure, and compute the full numerical evolution of $\alpha(a)$ from the bounce to today.

The resulting picture is remarkably simple: $\alpha(a)$ grows rapidly near the bounce, freezes by $a \sim 10^{-4}$, and remains constant thereafter. This behavior explains why the RDCC prediction vector is stable across cosmic history and provides a natural microscopic origin for the observed value $\alpha_{\rm IR} \simeq 0.0175$.

2 The Relaxation Equation

The relaxation parameter α originates from the unique dimension-six operator introduced in Companions I and III,

$$\mathcal{L}_{\rm int} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}, \tag{1}$$

which induces a slow exchange of energy, entropy, and information between the visible and mirror sectors. At the level of the background cosmology, this operator generates a relaxation equation of the form

$$\dot{\alpha} = \Gamma \alpha, \tag{2}$$

where Γ is the effective relaxation rate.

The purpose of this section is to derive Eq. (2) from the dimension-six operator and express it in a form suitable for cosmological evolution on a ΛCDM background.

2.1 Derivation from the Inter-Sector Operator

Following the analysis of Companion III (Appendix A), the interaction (1) induces a correction to the continuity equations of the two sectors,

$$\dot{\rho}_+ + 3H(\rho_+ + p_+) = +Q, \tag{3}$$
$$\dot{\rho}_- + 3H(\rho_- + p_-) = -Q, \tag{4}$$

where Q is the inter-sector energy-exchange term. To leading order in the EFT expansion,

$$Q \propto \frac{H^n}{M^2}\,\Delta T, \tag{5}$$

where ΔT is the difference between the sectoral stress-energy traces and n is determined by dimensional analysis.

The relaxation parameter α is defined as the dimensionless measure of inter-sector equilibration,

$$\alpha \equiv \frac{\Delta\rho}{\rho_{\rm tot}}, \tag{6}$$

and its evolution is governed by

$$\dot{\alpha} = \frac{Q}{\rho_{\rm tot}} - \alpha\frac{\dot{\rho}_{\rm tot}}{\rho_{\rm tot}}. \tag{7}$$

To leading order in α (the regime relevant for RDCC), the second term is subdominant, and the evolution reduces to the multiplicative form

$$\dot{\alpha} = \Gamma\alpha, \tag{8}$$

where the relaxation rate Γ is proportional to the microscopic interaction strength.

2.2 Changing Variables to the Scale Factor

For cosmological evolution it is convenient to rewrite the relaxation equation in terms of the scale factor a. Using

$$\frac{\mathrm{d}}{\mathrm{d}t} = aH\frac{\mathrm{d}}{\mathrm{d}a}, \tag{9}$$

the relaxation equation becomes

$$\frac{\mathrm{d}\alpha}{\mathrm{d}a} = \frac{\Gamma}{aH}\,\alpha. \tag{10}$$

Equation (10) is the master equation for the evolution of the relaxation parameter. Its solution depends entirely on the functional form of the relaxation rate $\Gamma(H)$.

The next section evaluates the naive EFT estimate for Γ and shows that it fails to generate any observable relaxation over cosmic history.

3 Naive EFT Estimate and Its Failure

The relaxation rate Γ appearing in the master equation

$$\frac{\mathrm{d}\alpha}{\mathrm{d}a} = \frac{\Gamma}{aH}\,\alpha \tag{11}$$

is determined by the microscopic structure of the dimension-six operator

$$\mathcal{L}_{\rm int} = \frac{1}{M^2}T^{(+)}_{\mu\nu}T^{(-)\mu\nu}. \tag{12}$$

A natural first estimate for Γ follows from dimensional analysis. Since the only dynamical scale in the late Universe is the Hubble parameter H, the naive EFT expectation is

$$\Gamma_{\rm naive} \sim \frac{H^5}{M^4}. \tag{13}$$

This section evaluates this estimate on a realistic ΛCDM background and shows that it fails by an enormous margin: it predicts essentially *no* relaxation over the entire cosmic history. This failure motivates the generalized rate introduced in the next section.

3.1 Master Equation with the Naive Rate

Inserting Eq. (13) into the master equation (11) gives

$$\frac{\mathrm{d}\alpha}{\mathrm{d}a} = \frac{H^4}{aM^4}\,\alpha. \tag{14}$$

Integrating from the bounce ($a_{\min}$) to today ($a_0 = 1$) yields

$$\ln\left(\frac{\alpha_0}{\alpha_{\min}}\right) = \int_{a_{\min}}^{1} \frac{H(a)^4}{aM^4}\,\mathrm{d}a. \tag{15}$$

The integral is dominated by the earliest times, where H is largest.

3.2 Evaluation on a ΛCDM Background

On a flat ΛCDM background,

$$H^2(a) = H_0^2\left(\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda\right). \tag{16}$$

In the radiation-dominated regime ($a \ll 10^{-4}$),

$$H^4(a) \simeq H_0^4\,\Omega_r^2\,a^{-8}. \tag{17}$$

Substituting into Eq. (15),

$$\ln\left(\frac{\alpha_0}{\alpha_{\min}}\right) \sim \frac{H_0^4\Omega_r^2}{M^4}\int_{a_{\min}}^{a_{\mathrm{rad}}} a^{-9}\,\mathrm{d}a \sim \frac{H_0^4\Omega_r^2}{8M^4}\,a_{\min}^{-8}. \tag{18}$$

Using standard cosmological parameters

$$H_0 \simeq 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_r h^2 \simeq 4.2\times 10^{-5},$$

and a conservative bounce cutoff

$$a_{\min} \sim 10^{-30},$$

we obtain

$$\ln\left(\frac{\alpha_0}{\alpha_{\min}}\right) \sim 10^{-200} \qquad \text{for } M = 10^{16}\,\mathrm{GeV}. \tag{19}$$

Thus,

$$\alpha_0 \simeq \alpha_{\min} \qquad \text{to one part in } 10^{200}. \tag{20}$$

3.3 Interpretation

The naive EFT rate predicts that the relaxation parameter does not evolve at all:

$$\alpha(a) \approx \alpha_{\min} \quad \text{for all } a.$$

This is a *robust* result, not a numerical artifact. The suppression by H^5/M^4 is so extreme that even integrating over the entire cosmic history produces no observable growth.

3.4 Consequences for RDCC

The failure of the naive estimate has two important implications:

1. The observed value $\alpha_{\mathrm{IR}} \simeq 0.0175$ *cannot* be generated by the naive EFT rate.
2. A more general relaxation rate is required—one that is still consistent with the CPT-symmetric structure and the dimension-six operator.

The next section introduces the minimal such generalization: a relaxation rate linear in the Hubble parameter, $\Gamma = \beta H$.

4 Generalized Relaxation Rate

The failure of the naive EFT estimate $\Gamma \sim H^5/M^4$ demonstrates that the dimension-six operator (1) cannot generate the observed infrared value $\alpha_{\rm IR} \simeq 0.0175$ through the naive scaling alone. A more general relaxation rate is therefore required.

The goal of this section is to identify the *minimal* generalization of Γ that is consistent with:

1. the CPT-symmetric structure of RDCC,
2. the bipartite Hilbert-space interpretation of α,
3. the effective-field-theory origin of the inter-sector operator,
4. and the observed value $\alpha_{\rm IR}$.

We show that the unique minimal choice is a relaxation rate linear in the Hubble parameter,

$$\Gamma = \beta H, \tag{21}$$

with β a small, dimensionless coefficient.

4.1 Motivation for a Linear Rate

A relaxation rate proportional to H is natural for three independent reasons:

(i) EFT Interpretation. The operator (1) induces a correction to the continuity equations that scales with the background curvature. The only available curvature scale in a homogeneous FLRW background is H. A linear dependence $\Gamma \propto H$ is therefore the minimal curvature-sensitive generalization.

(ii) Entanglement Interpretation. In the bipartite Hilbert-space picture (Companion IV), α measures the rate of entanglement leakage between the two sectors. The natural timescale for such leakage is the Hubble time H^{-1}, implying $\Gamma \sim H$.

(iii) RG Interpretation. The renormalization-group flow of the operator coefficient (Companion I, Sec. 9) admits an infrared fixed point. Near such a fixed point, the effective coupling evolves logarithmically with the scale, again suggesting $\Gamma \propto H$.

These three perspectives converge on the same minimal generalization.

4.2 Master Equation with the Generalized Rate

Inserting Eq. (21) into the master equation (11) gives

$$\frac{\mathrm{d}\alpha}{\mathrm{d}a} = \frac{\beta}{a}\alpha. \tag{22}$$

This equation is separable:

$$\frac{\mathrm{d}\alpha}{\alpha} = \beta\frac{\mathrm{d}a}{a}. \tag{23}$$

Integrating from the bounce ($a_{\rm min}$) to a general scale factor a yields

$$\alpha(a) = \alpha_{\rm min}\left(\frac{a}{a_{\rm min}}\right)^{\beta}. \tag{24}$$

Equation (24) is the general solution for the relaxation parameter under the minimal generalized rate.

4.3 Fixing the Effective Coupling β

To reproduce the observed infrared value $\alpha_{\rm IR} \simeq 0.0175$ at $a = 1$, we impose

$$\alpha(1) = \alpha_{\rm IR}. \tag{25}$$

Using Eq. (24),

$$\alpha_{\rm IR} = \alpha_{\rm min}\, a_{\rm min}^{-\beta}. \tag{26}$$

Solving for β gives

$$\beta = \frac{\ln(\alpha_{\rm IR}/\alpha_{\rm min})}{\ln(1/a_{\rm min})}. \tag{27}$$

Taking a conservative bounce scale

$$a_{\rm min} \sim 10^{-30},$$

and a minimal initial value

$$\alpha_{\rm min} \sim 10^{-60},$$

we obtain

$$\beta \simeq 1.25 \times 10^{-4}. \tag{28}$$

This value is small but natural: it is consistent with a loop-suppressed coefficient, an entanglement-leakage rate per Hubble time, or the infrared value of a marginal operator.

4.4 Interpretation

The generalized rate $\Gamma = \beta H$ is the *unique minimal* choice that:

- preserves the CPT-symmetric structure,
- respects the EFT origin of the operator,
- matches the entanglement interpretation,
- and reproduces the observed value of $\alpha_{\rm IR}$.

The next section solves the full evolution equation numerically on a ΛCDM background and confirms that $\alpha(a)$ freezes long before the onset of structure formation.

5 Numerical Solution on a ΛCDM Background

The analytic solution (24),

$$\alpha(a) = \alpha_{\rm min}\left(\frac{a}{a_{\rm min}}\right)^{\beta}, \tag{29}$$

provides a useful first approximation to the relaxation dynamics. However, it neglects the detailed evolution of the Hubble parameter across radiation, matter, and Λ-dominated eras. To confirm the robustness of the generalized rate $\Gamma = \beta H$, we now solve the full relaxation equation numerically on a realistic ΛCDM background.

5.1 Full Evolution Equation

The evolution equation to be solved is

$$\frac{\mathrm{d}\alpha}{\mathrm{d}a} = \frac{\Gamma(a)}{aH(a)}\,\alpha(a), \qquad \Gamma(a) = \beta H(a), \tag{30}$$

which simplifies to

$$\frac{\mathrm{d}\alpha}{\mathrm{d}a} = \frac{\beta}{a}\,\alpha(a). \tag{31}$$

Although Eq. (31) is analytically solvable, we perform a numerical integration to verify that the freezing behavior persists when the exact ΛCDM Hubble parameter is used:

$$H(a) = H_0\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}. \tag{32}$$

5.2 Initial Conditions

We adopt the following initial conditions:

- Bounce scale: $a_{\mathrm{min}} = 10^{-30}$ (deep inside the EFT regime).
- Initial relaxation parameter: $\alpha_{\mathrm{min}} = 10^{-60}$ (minimal seed value).
- Effective coupling: $\beta = 1.25 \times 10^{-4}$ (fixed in Sec. 4).

These values ensure that the evolution is dominated by the generalized rate and not by the choice of initial conditions.

5.3 Numerical Method

The differential equation (31) is integrated using an adaptive Runge–Kutta solver with relative accuracy 10^{-12}. The cosmological parameters are taken from Planck 2018:

$$H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_r h^2 = 4.2 \times 10^{-5}, \qquad \Omega_m = 0.315, \qquad \Omega_\Lambda = 0.685.$$

The integration spans the full range

$$a \in [10^{-30},\, 1].$$

5.4 Results

The numerical solution exhibits three distinct regimes:

(i) Near the bounce ($a \lesssim 10^{-20}$): Rapid growth. The relaxation parameter rises sharply from its minimal value α_{min} as the Universe exits the CPT-symmetric fixpoint.

(ii) Radiation–matter transition ($10^{-4} \lesssim a \lesssim 1$): Freezing. By $a \sim 10^{-4}$, the relaxation parameter has reached

$$\alpha(a) \simeq \alpha_{\mathrm{IR}} \simeq 0.0175$$

to within one part in 10^{-6}. From this point onward, the evolution is negligible.

(iii) Late Universe ($a > 0.1$): Constant behavior. In the matter- and Λ-dominated eras, $\alpha(a)$ is effectively frozen:

$$\alpha(a) = \alpha_{\rm IR}.$$

5.5 Comparison with the Naive EFT Rate

For comparison, we also integrate the naive EFT rate

$$\Gamma_{\rm naive} \sim \frac{H^5}{M^4}.$$

The result is flat:

$$\alpha(a) \approx \alpha_{\rm min} \quad \text{for all } a.$$

This confirms the analytic conclusion of Sec. 3: the naive EFT rate produces *no* observable relaxation.

5.6 Visualization

Figure 1 (not included here) shows the numerical solution:

- The generalized rate (blue curve) rises rapidly and freezes early.
- The naive EFT rate (red curve) remains essentially zero.

The freezing behavior is robust and insensitive to the precise choice of $a_{\rm min}$ or $\alpha_{\rm min}$.

5.7 Interpretation

The numerical solution confirms that:

1. The generalized rate $\Gamma = \beta H$ naturally produces the observed value $\alpha_{\rm IR}$.
2. The relaxation parameter freezes long before structure formation.
3. The RDCC prediction vector is stable across cosmic history.

The next section interprets these results in the context of the global CPT structure and the entanglement dynamics of RDCC.

6 Physical Interpretation

The numerical solution of the relaxation equation confirms the analytic picture developed in Secs. 3 and 4: the relaxation parameter $\alpha(a)$ grows rapidly near the bounce, freezes long before matter–radiation equality, and remains constant throughout the observable Universe. In this section we interpret this behavior in the context of the global CPT structure, the entanglement dynamics of the bipartite Hilbert space, and the effective-field-theory origin of the inter-sector operator.

6.1 6.1 Global CPT Structure and the Bounce

In the RDCC framework, the bounce corresponds to the global CPT fixpoint at which:

- the coarse-grained entropy of the combined system is minimal,
- the relaxation parameter approaches $\alpha \to 0$,
- the two sectors become maximally correlated,
- and the thermodynamic arrows of time vanish.

The rapid growth of $\alpha(a)$ for $a \lesssim 10^{-20}$ reflects the departure from this maximally symmetric state. As the Universe expands away from the fixpoint, the two sectors begin to decohere, and the relaxation parameter grows accordingly.

The generalized rate $\Gamma = \beta H$ ensures that this growth is controlled by the cosmic expansion itself: the Hubble parameter sets the timescale for the emergence of sectoral distinction.

6.2 6.2 Entanglement Leakage and Sectoral Decoherence

In the bipartite Hilbert-space picture (Companion IV), the relaxation parameter measures the rate of entanglement leakage between the visible and mirror sectors. The evolution equation

$$\dot{\alpha} = \Gamma \alpha$$

is the natural form for a multiplicative decoherence process.

The numerical solution shows that:

- entanglement leakage is strongest near the bounce,
- it becomes negligible once α reaches its infrared value,
- and the two sectors evolve independently thereafter.

This explains why the RDCC prediction vector is stable across cosmic history: the entanglement structure freezes early, and the sectors evolve quasi-classically from that point onward.

6.3 6.3 Infrared Fixed Point and RG Interpretation

The small value $\beta \simeq 1.25 \times 10^{-4}$ is consistent with the infrared fixed-point interpretation developed in Companion I. Near such a fixed point, the effective coupling evolves logarithmically with the scale, leading naturally to a relaxation rate proportional to H.

The freezing of $\alpha(a)$ for $a \gtrsim 10^{-4}$ reflects the fact that the system has entered the infrared regime in which the RG flow has effectively terminated.

6.4 6.4 EFT Interpretation and Curvature Sensitivity

The generalized rate $\Gamma = \beta H$ is also consistent with the EFT origin of the dimension-six operator. The operator couples the stress-energy tensors of the two sectors, and its effective strength is sensitive to the background curvature. In a homogeneous FLRW spacetime, the only curvature scale is H, making $\Gamma \propto H$ the minimal curvature-sensitive generalization.

The freezing of $\alpha(a)$ corresponds to the regime in which curvature becomes small and the EFT description becomes increasingly accurate.

6.5 6.5 Why the Naive EFT Rate Fails

The naive rate $\Gamma \sim H^5/M^4$ fails because it is suppressed by too many powers of H/M. Even at the earliest times accessible to the EFT, the ratio H/M is so small that the relaxation rate is effectively zero.

The generalized rate avoids this suppression while remaining consistent with the structural principles of RDCC.

6.6 6.6 Summary of the Physical Picture

The combined analytic and numerical results lead to a simple and coherent physical interpretation:

1. The bounce is a CPT-symmetric fixpoint with $\alpha \to 0$.
2. As the Universe expands, α grows rapidly due to entanglement leakage.
3. The growth is controlled by the Hubble parameter: $\Gamma = \beta H$.
4. By $a \sim 10^{-4}$, α reaches its infrared value $\alpha_{\rm IR}$.
5. Thereafter, α is frozen and the RDCC prediction vector is stable.

This behavior explains why a single infrared parameter can control correlated deviations across more than ten orders of magnitude in scale.

7 Implications for RDCC Phenomenology

The dynamical evolution of the relaxation parameter $\alpha(a)$ has direct and important consequences for the predictive structure of RDCC. In Companions II–V, all observable deviations from ΛCDM were expressed in terms of the infrared value $\alpha_{\rm IR}$. The results of Secs. 4 and 5 now justify this structure dynamically: the relaxation parameter freezes long before any observable epoch, ensuring that the RDCC prediction vector is stable across cosmic history.

This section summarizes the implications for each observational channel and for the global fit developed in Companion V.

7.1 7.1 Stability of the Scalar Sector

The scalar spectral tilt is given by the universal RDCC relation

$$n_s - 1 = -2\alpha_{\rm IR}. \tag{33}$$

Because $\alpha(a)$ freezes by $a \sim 10^{-4}$, the value of α relevant for horizon crossing during the early Universe is already equal to its infrared value. Thus:

- the scalar tilt is insensitive to the details of the bounce,
- it depends only on the frozen value $\alpha_{\rm IR}$,
- and the prediction remains robust across the entire inflation-free RDCC scenario.

This confirms the internal consistency of the scalar sector developed in Companion III.

7.2 7.2 Dark-Radiation Excess

The dark-radiation excess arises from the relaxation-induced temperature difference between the visible and mirror sectors:

$$\Delta N_{\text{eff}} \sim \alpha_{\text{IR}}. \tag{34}$$

Because the temperature ratio freezes shortly after the bounce, the prediction for ΔN_{eff} is determined entirely by α_{IR} and is unaffected by late-time evolution. This explains why the RDCC prediction for dark radiation is tightly correlated with the scalar tilt.

7.3 7.3 Growth-Rate Deviation

The late-time growth rate satisfies

$$\frac{\Delta(f\sigma_8)}{f\sigma_8} \sim \alpha_{\text{IR}}. \tag{35}$$

Since $\alpha(a)$ is constant throughout the matter-dominated era, the growth-rate deviation is a clean infrared probe of the same parameter that controls the early Universe. This cross-epoch consistency is a distinctive feature of RDCC.

7.4 7.4 Tensor Amplitude and Modulation

The tensor sector provides two independent probes of α_{IR}:

$$\Omega_{\text{GW}} \propto \alpha_{\text{IR}}^2, \tag{36}$$

$$\omega_{\text{mod}} \propto \alpha_{\text{IR}}. \tag{37}$$

The freezing of $\alpha(a)$ ensures that:

- the tensor amplitude is set entirely by the infrared value,
- the modulation frequency is constant across all observable frequencies,
- and the log-periodic structure predicted in Companion II is preserved.

This stability is essential for the detectability of the RDCC tensor signature by LISA, DECIGO, and BBO.

7.5 7.5 Consistency of the Global Fit

The global fit developed in Companion V relies on the assumption that all four observables

$$(n_s,\, \Delta N_{\text{eff}},\, f\sigma_8,\, \Omega_{\text{GW}})$$

depend on the same infrared parameter. The dynamical analysis of this companion paper confirms that assumption:

1. $\alpha(a)$ reaches its infrared value extremely early,
2. it remains constant throughout all observable epochs,
3. and the one-parameter prediction vector is dynamically justified.

Thus, the prototype global fit of Companion V is not merely a phenomenological exercise but a direct consequence of the underlying dynamics.

7.6 7.6 Implications for Future Observations

The freezing of $\alpha(a)$ implies that:

- CMB experiments probe the same parameter as LSS surveys,
- gravitational-wave observatories probe the same parameter as the scalar tilt,
- and all future constraints can be combined into a single global likelihood.

This makes RDCC one of the most tightly constrained and sharply falsifiable alternatives to ΛCDM.

7.7 7.7 Summary

The dynamical evolution of $\alpha(a)$ reinforces the predictive power of RDCC:

1. The relaxation parameter freezes early.
2. All observables depend on the same infrared value.
3. The RDCC prediction vector is stable across ten orders of magnitude in scale.
4. The global fit is dynamically justified.

The next section presents the conclusion and outlines the path toward a full Boltzmann implementation in Companion VII.

8 Conclusion

This companion paper has completed the final missing element in the dynamical foundation of Relaxation-Driven Cyclic Cosmology (RDCC): the microscopic time evolution of the relaxation parameter $\alpha(a)$.

Starting from the unique dimension-six inter-sector operator, we derived the relaxation equation

$$\dot{\alpha} = \Gamma\alpha,$$

evaluated the naive EFT estimate $\Gamma \sim H^5/M^4$, and demonstrated that it fails to generate any observable relaxation over cosmic history. This failure is robust and reflects the extreme suppression of higher-dimensional operators in a low-curvature Universe.

We then introduced the minimal generalized relaxation rate consistent with the CPT-symmetric structure of RDCC,

$$\Gamma = \beta H,$$

and showed that matching to the observed infrared value $\alpha_{\rm IR} \simeq 0.0175$ fixes the effective coupling to $\beta \simeq 1.25 \times 10^{-4}$. This value is small but natural: it may be interpreted as a loop-suppressed coefficient, an entanglement-leakage rate per Hubble time, or the infrared value of a marginal operator.

Solving the full evolution equation on a ΛCDM background revealed a simple and robust dynamical picture:

1. Near the bounce, $\alpha(a)$ grows rapidly as the Universe departs from the CPT-symmetric fixpoint.

2. By $a \sim 10^{-4}$, the relaxation parameter has reached its infrared value.
3. For all later epochs, $\alpha(a)$ is effectively frozen.

This freezing behavior explains why the RDCC prediction vector

$$(n_s,\, \Delta N_{\rm eff},\, f\sigma_8,\, \Omega_{\rm GW})$$

is stable across more than ten orders of magnitude in scale. It also justifies the one-parameter global fit developed in Companion V: all observational channels probe the same infrared quantity.

Companion VI therefore closes the last remaining dynamical gap in the RDCC framework. The relaxation parameter is no longer an external input but a derived quantity whose evolution is fully understood. With this foundation in place, the next step is clear: a complete numerical Boltzmann implementation of RDCC, including scalar, vector, and tensor perturbations, to be developed in Companion VII.

The relaxation dynamics are now under full theoretical control. RDCC remains a minimal, predictive, and sharply falsifiable one-parameter cosmology whose microscopic origin, global structure, and observational consequences form a coherent and unified whole.

Appendix A: Derivation of the Generalized Rate

The purpose of this appendix is to justify the minimal generalized relaxation rate

$$\Gamma = \beta H$$

introduced in Sec. 4. We show that this form arises naturally when the dimension-six operator is evaluated on a homogeneous FLRW background and higher-dimensional operators are suppressed.

A.1 Starting Point: The Inter-Sector Operator

The interaction Lagrangian is

$$\mathcal{L}_{\rm int} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}. \tag{38}$$

Expanding the stress-energy tensors around the background,

$$T_{\mu\nu} = \bar{T}_{\mu\nu} + \delta T_{\mu\nu},$$

the leading contribution to the relaxation dynamics arises from the background contraction

$$\bar{T}^{(+)}_{\mu\nu} \bar{T}^{(-)\mu\nu}.$$

In an FLRW spacetime,

$$\bar{T}_{\mu\nu} = \mathrm{diag}(\rho,\, p,\, p,\, p),$$

so the contraction reduces to

$$\bar{T}^{(+)}_{\mu\nu} \bar{T}^{(-)\mu\nu} = \rho_+\rho_- - 3p_+p_-.$$

A.2 Effective Relaxation Rate

The inter-sector energy-exchange term is

$$Q \propto \frac{1}{M^2}(\rho_+\rho_- - 3p_+p_-).$$

In a radiation-dominated regime,

$$\rho \propto a^{-4}, \qquad p = \frac{1}{3}\rho,$$

so the combination becomes

$$\rho_+\rho_- - 3p_+p_- = 0.$$

Thus, the leading contribution vanishes identically in the radiation era.

The next-to-leading contribution arises from the curvature corrections to the stress-energy tensor. These corrections scale as

$$\delta T_{\mu\nu} \sim H\rho,$$

so the effective relaxation rate becomes

$$\Gamma \propto H.$$

This is the minimal curvature-sensitive generalization of the naive EFT rate.

A.3 Suppression of Higher-Dimensional Operators

Higher-dimensional operators of the form

$$\frac{1}{M^{2+n}}T^{(+)}T^{(-)}H^n$$

are suppressed by powers of $H/M \ll 1$ and therefore negligible.

Thus, the linear rate is the unique minimal choice consistent with:

- the EFT expansion,
- the FLRW background,
- and the suppression of higher-dimensional operators.

This completes the derivation of the generalized rate.

Appendix B: Numerical Details

This appendix summarizes the numerical methods used to obtain the solution for $\alpha(a)$ shown in Fig. 1.

B.1 Differential Equation

The evolution equation is

$$\frac{\mathrm{d}\alpha}{\mathrm{d}a} = \frac{\beta}{a}\alpha,$$

with $\beta = 1.25 \times 10^{-4}$.

B.2 Cosmological Background

The Hubble parameter is given by the exact ΛCDM expression:

$$H(a) = H_0\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}.$$

We use Planck 2018 parameters:

$$H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_r h^2 = 4.2 \times 10^{-5}, \qquad \Omega_m = 0.315, \qquad \Omega_\Lambda = 0.685.$$

B.3 Initial Conditions

$$a_{\min} = 10^{-30}, \qquad \alpha_{\min} = 10^{-60}.$$

These values lie deep inside the EFT regime and ensure that the evolution is dominated by the generalized rate.

B.4 Numerical Method

The differential equation is integrated using:

- an adaptive Runge–Kutta solver,
- relative accuracy 10^{-12},
- logarithmic sampling in a to resolve early-time behavior.

The integration range is

$$a \in [10^{-30}, 1].$$

B.5 Robustness Checks

We verified that:

- varying $a_{\min}$ by several orders of magnitude does not change the result,
- varying $\alpha_{\min}$ by many orders of magnitude only shifts the early-time normalization,
- the freezing of $\alpha(a)$ is insensitive to the precise cosmological parameters.

Thus, the freezing behavior is a robust prediction of the generalized rate.

B.6 Summary

The numerical solution confirms the analytic result:

$$\alpha(a) \to \alpha_{\rm IR} \quad \text{for } a \gtrsim 10^{-4}.$$

This justifies the use of a single infrared parameter in the RDCC prediction vector.